

MRI Foreground/Background Detection

Amir R. Sadri

ars329@case.edu

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Abstract

This note outlines how we derive foreground and background masks for MRI volumes in RadQy. Each section documents one detection method, its assumptions, and the clean-up steps needed so FG/BG pixels are reproducible across datasets.

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1 Introduction

This document lists the foreground/background (FG/BG) segmentation methods in `backend/seg/mri`. Each section mirrors a Python module and notes its inputs, thresholds, morphology, and fallbacks so downstream IQMs get consistent masks.

2 Image Processing Segmenters

Classical pipelines based on intensity statistics and morphology.

2.1 OtsuHull (`OtsuHull.py`)

Legacy variant mirroring the original hybrid Otsu pipeline with more defensive fallbacks [1]. This subsection is listed first because it matches the earliest implementation shipped with RadQy.

Algorithm 1: OtsuHull foreground/background estimation

1. Input slice I ; cast to uint16.
2. Histogram-equalize I to get H .
3. Otsu on I and H to obtain masks m_1, m_2 and FG fractions w_1, w_2 .
4. Hybrid recombination: $S = w_1 \cdot I + w_2 \cdot H$ (normalize to image range).
5. Otsu on S to get final mask M .
6. Cleanup: convex hull of M (fallback to hole filling if hull fails).
7. Outputs: $fg = M, bg = 1 - M$.

Notes on the weights and masks: m_1 is the Otsu mask on the raw slice I , m_2 is the Otsu mask on the hist-equalized slice H . The weights $w_1 = |m_1|/|I|$ and $w_2 = |m_2|/|I|$ measure how much of each slice is classified as foreground; they scale the raw and equalized images when forming $S = w_1 \cdot I + w_2 \cdot H$ so that the cleaner mask (smaller foreground fraction) influences the recombined image more.

2.2 Adaptive Border Segmentation (`AdaptiveBorderSeg.py`)

Single-slice segmenter that estimates background from the image border and keeps the largest non-background component.

Algorithm 2: AdaptiveBorder foreground/background estimation

1. Normalize input slice to uint8 via 1st/99th percentile scaling.
2. Extract border ring (default 12 px each side); compute μ, σ on non-zero border pixels (fallback: full image if border empty).
3. Mark background where $I < \mu - 1\sigma$.
4. Morphology: binary closing then opening with 5×5 kernel; fill holes.
5. Foreground = inverse of background; keep largest connected component.
6. Safety: if FG area $< 5\%$, set FG to all-ones (entire slice).

7. Outputs: (fg, bg) with $bg = 1 - fg$.

2.3 Hybrid Otsu + Convex Hull (HybridOtsuHullSeg.py)

Modern hybrid approach combining Otsu on raw and histogram-equalized images with convex hull cleanup.

Algorithm 3: Hybrid Otsu + Convex Hull

1. Robust-scale input slice to $[0, 1]$ via 1st/99th percentiles; cast to uint8 for thresholding.
2. Compute Otsu masks on raw (x) and hist-equalized (h) images to get FG fractions w_1, w_2 .
3. Recombine: $S = w_1 \cdot x + w_2 \cdot h$.
4. Otsu on S to obtain mask M .
5. Cleanup: apply convex hull to M (fallback: hole filling if hull unavailable).
6. Outputs: (fg, bg) with $fg = M$, $bg = 1 - fg$.

3 Deep Learning Segmenters

Learned models for FG/BG inference.

3.1 U-Net (placeholder)

Planned slot for the learned U-Net FG/BG mask generator.

Algorithm 4: U-Net inference pipeline (placeholder)

1. Preprocess: normalize/standardize input slice to network expectations.
2. Forward pass through U-Net to obtain probability map P .
3. Threshold P (e.g., at 0.5) to get binary mask M .
4. Post-process: optional hole filling, small-component removal, and convex hull if needed.
5. Outputs: (fg, bg) with $fg = M$, $bg = 1 - fg$.

4 Notes for Integration

All segmenters operate per 2D slice and return uint8 masks in $\{0, 1\}$. Background is always defined as the complement of foreground to ensure downstream IQMs receive matching shapes and disjoint FG/BG sets.

References

- [1] Amir Reza Sadri et al. “MRQy: An open-source tool for the quality control of structural MRI images”. In: *Journal of Imaging* 6.1 (2020). Placeholder entry; replace with canonical citation if available, pp. 1–15.